

Nithin K R

nithinpoojari717@gmail.com • [+91-9019971722](tel:+91-9019971722) • linkedin.com/in/nithinkr06
• github.com/NITHINKR06

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Dear Hiring Manager,

I am a final-year B.Tech student specialising in **Computer Science & Cyber Security** at NMAM Institute of Technology, Nitte, with a diploma background in full-stack web development and over two years of hands-on project experience spanning security engineering, AI/ML pipelines, and production-grade web applications. I am actively seeking an opportunity where I can contribute immediately as a **Full-Stack Engineer**, **Security Engineer**, or in a hybrid role that sits at the intersection of both disciplines. What sets me apart is not just the range of my technical skills, but my ability to ship complete, working systems independently from first principles to production deployment.

Who I Am

My engineering philosophy is simple: *build things that work in the real world, not just on a laptop*. Every project I have undertaken has been shipped end-to-end from architecture decisions and database schemas, through REST and LLM APIs, to containerised deployments and polished frontends without relying on a team. This solo ownership has forced me to develop a practical grasp of cross-cutting concerns that are often siloed in larger teams: authentication flows, caching strategies, rate limiting, CI/CD pipelines, and threat modelling. I have learned to make pragmatic engineering decisions under real constraints not just the ideal-world choices that look good on a whiteboard.

Beyond code, I serve as **Student Technical Lead** for the CSI chapter at NMAM (500+ members), where I plan and deliver technical workshops, mentor junior developers on real projects, and translate complex systems concepts into accessible language. This role has sharpened both my technical communication and my ability to take ownership of outcomes skills I consider as important as any language or framework.

What I Have Built

The four projects below represent independent, end-to-end work not course assignments or tutorials, but systems I architected, built, and deployed myself:

- **CodeSentinel** (Full-Stack Security Analysis Platform)
Built with **Next.js**, **FastAPI**, and **Docker Compose**, CodeSentinel is a complete static-analysis and exploit-simulation platform. It scans GitHub repositories and live URLs using AST-based pattern matching across 12+ vulnerability types, detects multi-step vulnerability chains, and generates working proof-of-concept exploits. A **Groq/Ollama-powered LLM patch generator** produces remediation diffs that are then validated by a red-agent loop before being automatically submitted as GitHub Pull Requests. All scan results and attack relationships are rendered in a **live D3.js attack-graph dashboard** with real-time updates. This project required me to reason deeply about adversarial systems, design LLM-in-the-loop safety workflows, and coordinate multiple services across a Docker Compose network skills directly transferable to any security or platform engineering role.
- **DriftGuard** (Ensemble ML Network Intrusion Detection System)
DriftGuard is an anomaly-detection NIDS that fuses three complementary model architectures **Isolation Forest** (density-based outlier detection), **Dense Autoencoder** (reconstruction-

error anomalies), and **Bidirectional LSTM** (temporal sequence modelling) using weighted ensemble fusion to combine their strengths. The system achieves $\sim 2,380$ flows/sec throughput and was rigorously validated on three real-world public datasets: CICIDS2017, UNSW-NB15, and UGR16v2. Beyond raw detection, it integrates **SHAP-based explainability** to produce human-readable reports that explain *why* a flow was flagged making it analyst-friendly rather than a black box. Building this taught me to make principled ML architecture decisions under real performance and interpretability constraints.

- **WALRUS** (Cybersecurity Education & Scam Detection Platform)

WALRUS is a production-hardened full-stack platform (TypeScript · React · Node.js · MongoDB) designed to raise cybersecurity literacy through gamified learning modules, daily streaks, and leaderboards. Its centrepiece is an **AI-powered scam analyser** using Hugging Face **BART-large-MNLI** to classify phishing URLs, suspicious emails, and phone numbers with calibrated confidence scores. The platform is fully multilingual (English, Hindi, Kannada) and security-hardened with Helmet.js, strict CORS policies, and aggressive rate limiting. Designing WALRUS required balancing user experience, model inference latency, and production security simultaneously a challenge that reflects real-world engineering constraints.

- **GitPulse** (High-Performance Async REST API)

GitPulse is a **Spring Boot** REST API (Java · Redis · Docker) that aggregates GitHub developer metrics across six endpoints. It uses **@Async + CompletableFuture** for fully non-blocking concurrent API calls and **Redis caching** with a 10-minute TTL to stay well within GitHub's rate limits while maintaining fast response times. Results are surfaced as dark-themed **Thymeleaf** profile cards. This project demonstrates my ability to work fluently in Java ecosystems and reason carefully about concurrency, caching trade-offs, and external API constraint management.

What I Bring to Your Team

- **Security depth, not just awareness** My NPTEL *Internet Crimes and Cyber Security* certification (**96 %**), Oracle Cloud Infrastructure Networking Professional credential, **Tata Cybersecurity Analyst Job Simulation** (Forage, Jun 2026), **Hedera Hashgraph Developer** credential (The Hashgraph Association, Oct 2025), and hands-on penetration testing background (Kali Linux, OWASP Top 10, CVSS scoring) mean I design for resilience from the first commit not as an afterthought. I understand attack surfaces, not just defences.
 - **Practical AI/ML integration** I have deployed Hugging Face transformers, GPT-4o, scikit-learn pipelines, TensorFlow/Keras models, SHAP explainability, and local LLMs (Ollama) in production contexts. I can make real trade-off decisions about model selection, inference latency, accuracy budgets, and when *not* to use ML skills that go well beyond running notebooks.
 - **Full-stack breadth with depth** I am equally comfortable designing a normalised PostgreSQL schema, writing a FastAPI microservice, wiring a React component to an LLM stream, writing Java concurrency code, and configuring Docker Compose networking. I contribute across the entire stack from day one without a ramp-up phase on unfamiliar layers.
 - **Ownership mindset** I have shipped every project without a team. I write the tickets, make the architecture calls, debug production issues at 2am, and own the outcome end-to-end. I thrive in environments that reward initiative, curiosity, and direct accountability over process.
 - **Communication and mentorship** Leading workshops and mentoring developers has trained me to explain technical decisions clearly to non-expert audiences, write documentation that people actually read, and give feedback that improves others without discouraging them.
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Core Tech Stack

Languages	TypeScript, JavaScript, Python, Java, SQL
Frontend	Next.js, React.js, Tailwind CSS, Framer Motion, Three.js
Backend	FastAPI, Node.js/Express, Spring Boot, PostgreSQL, MongoDB
Security	Kali Linux, iptables, HMAC-SHA256, OWASP Top 10, CVSS
AI / ML	Hugging Face, GPT-4o, Ollama, scikit-learn, TensorFlow, SHAP
Infra	Docker, Redis, AWS, Vercel, Railway, GitHub Actions, Git

I am looking for a role where I can contribute meaningfully from the start whether that means hardening a system against attack, shipping a new product feature, building an internal tool, or integrating an LLM into a workflow that genuinely improves user outcomes. I am a fast learner, a self-starter, and someone who takes quality personally. I do not need to be told twice, and I do not stop at *good enough*.

I am available for an interview at your earliest convenience and happy to provide **live demos** of any project above, walk through the full codebase, pair on a problem, or complete a technical assessment in whatever format you prefer.

Thank you sincerely for your time and consideration.

Warm regards,

Nithin K R

B.Tech CSE (Cyber Security) NMAM Institute of Technology, Nitte
nithinpoojari717@gmail.com · +91-9019971722 · [linkedin.com/in/nithinkr06](https://www.linkedin.com/in/nithinkr06)